Are Tone Marks the Weak Point of Open Thai OCR? A Locked-Split Study of Eight Models

Thanakrit Samoenar, Thanawat Suriya, and Asst. Prof. Nont Kanungsukkasem, Ph.D.
School of Information Technology, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand
67070060@kmitl.ac.th, 67070236@kmitl.ac.th, nont@it.kmitl.ac.th

***Abstract*—Open vision-language models (VLMs) for Thai OCR now read most of a page correctly, but Thai vowels and tone marks are small, and one mark can change a word. This paper asks whether these marks are the weak point of open Thai OCR models, and where they are lost. We introduce a mark score that ignores reading order, penalizes surplus text, and gives no credit to the copies produced by a repetition loop. We develop the analysis on a calibration split of ThaiOCRBench and confirm it on a locked split of 352 items, read once, with eight open models. The answer is mostly no: in all eight models, text that is absent from the output costs 4.7 to 15.8 times more marks than a mark misread after a correctly read consonant. Given the correct preceding text, Typhoon OCR 1.5 (2B) and Pathumma choose the right tone mark about 99% of the time, with little help from the image. Qwen3-VL-4B knows the marks as well but misreads some of them when it writes the page itself, so the finding holds only in part for general models. Stopping at a detected repetition loop raised mark F1 in 12 of the 13 model-and-task cells where a model loops on about 5% of pages or more, by 4.8 to 35.6 points, and it changed 1 of 2,400 outputs that had ended on their own. The one cell that did not pass, Pathumma on full pages, rarely loops. A repetition penalty lowers F1 on pages that do not loop, and subtracting the language prior removes correct tone marks. All results come from one benchmark. Typhoon comes from the group that released it, and Pathumma's training data are not named, so we do not rank the two models.**

***Index Terms*—Thai OCR, vision-language models, diacritics, repetition loops, training-free decoding**

I. Introduction

Thai is written on a line of consonants, with vowel signs and tone marks stacked above and below it. We call these combining characters *marks*: upper vowels, lower vowels and tone marks (listed in Section III). A single mark can change the meaning and the sound of a word, as in ข่าว (news) and ข้าว (rice), so a transcription that drops or swaps a mark is wrong for the reader even when almost every other character is right. Yet Thai OCR is usually scored with whole-text measures, such as n-gram overlap and normalized edit distance in ThaiOCRBench [1], or character error rate in other work. These measures do not show whether marks are the problem, and sequence-level scores also penalize a model that reads the columns of a page in a different order from the reference.

Open vision-language models (VLMs) for Thai OCR are now available [2]–[4]. When one of them gets a mark wrong, there are three possible explanations. The visual evidence for a small mark may be weak. The language prior may override the evidence. Or the mark may be missing from the output altogether, because the model skipped the line, failed to find the text, or fell into a repetition loop. Each explanation calls for a different remedy: re-reading or zooming for weak evidence, subtracting the prior as contrastive decoding does [5], [6] for an overriding prior, and changing the decoding or locating the text for missing text. The output alone cannot tell us which of these happened inside the model. We therefore say that a line is *absent from the output*, and we do not claim that the model failed to see it.

We study this question on ThaiOCRBench with Typhoon OCR 1.5 and seven other open checkpoints, on the full-page OCR and text recognition tasks. We develop the analysis on a calibration split and then confirm it on a locked split that we read once. We make three contributions.

1) A mark score that ignores reading order (Section III). It scores upper vowels, lower vowels and tone marks line by line, wherever the line appears in the output. Surplus text is penalized, and copies produced by a repetition loop earn no credit.
2) A diagnosis of where marks are lost (Sections IV and VI). On the locked split, a line that is absent from the output costs 4.7 to 15.8 times more marks than a mark misread after a correctly read consonant, in all eight models. Given the correct preceding text, Typhoon and Pathumma choose the right tone mark about 99% of the time, with little help from the image. Qwen3-VL-4B shows that this does not hold for every model: it knows the marks as well, but it also misreads some of them while it writes the page.
3) An evaluation of training-free decoding (Sections V and VI). On the locked split, stopping at a detected loop raised mark F1 in 12 of the 13 registered cells and changed 1 of 2,400 outputs that had ended on their own. A repetition penalty lowers F1 on pages that do not loop, and subtracting the language prior removes correct tone marks.

For the models we studied, most of the marks that a Thai OCR model loses are lost before the mark is reached. Decoding and locating the text therefore matter more than seeing a tone mark more sharply.

II. Related Work

Thai OCR with VLMs. ThaiOCRBench [1] evaluates proprietary and open VLMs on Thai document tasks, including full-page OCR and text recognition. It uses n-gram overlap and edit-distance measures: BLEU, METEOR, F1 and normalized Levenshtein similarity, with ANLS for text recognition. Its introduction lists stacked diacritics among the features of Thai that English-centric datasets do not capture, but no score isolates the marks. Several open Thai OCR models have appeared since. Typhoon OCR [2], in version 1.5, is a full fine-tune of Qwen3-VL-2B [7], and its authors name infographics as an area for improvement. Pathumma-llm-vision 3.0 [3] is another Qwen3-VL-2B fine-tune. wayu-paxa-ocr-zero [4] fine-tunes PaddleOCR-VL [8] on synthetic Thai pages. Each reports aggregate accuracy; we ask where the marks are lost.

Diacritics in OCR. OCR is known to be difficult for scripts with many diacritics. Post-OCR correction has been proposed for Vietnamese [9], and an evaluation of Arabic OCR treats diacritized text as a separate condition [10]. We found no work that scores Thai marks separately from reading order and surplus text.

Repetition and omission. Decoding that maximizes likelihood produces repetitive text [11], and document OCR models show the same behavior. Nougat reports pages on which the model repeats a sentence, and it stops generation when a logit-variance signal falls [12]. olmOCR retries at a higher temperature [13]. The PaddleOCR-VL pipeline truncates repetitive output after decoding [14], and the released DeepSeek-OCR code bans repeated n-grams within a window [15], [16]. CC-OCR counts repetition as a form of OCR hallucination [17]. These works treat repetition as something to detect or avoid. We measure what each remedy costs the marks on pages that do not loop. One such remedy is the repetition penalty of CTRL [18], which some OCR packages enable by default.

Language priors and training-free decoding. Training-free decoding methods reduce hallucination in VLMs by favoring what the image supports over what the language prior alone would predict. VCD [5] contrasts the output distributions for the original and a distorted image, M3ID [6] amplifies the influence of the image over the language prior, and PAI [19] amplifies attention to image tokens and subtracts the logits of a text-only input. These methods assume that the language prior is the problem. For Thai tone marks we find that the prior is usually right. Our versions are reimplementations, so we call them M3ID-form and PAI-style. Speculative decoding [20] with prompt-lookup drafts [21] speeds up decoding and leaves the output unchanged in exact arithmetic.

III. Mark Score and Setup

We call the following characters marks: the six upper vowels (ั ิ ี ึ ื ็), the two lower vowels (ุ ู) and the four tone marks (่ ้ ๊ ๋). A full page has several reference lines, and a model may read its columns in a different order from the reference. A global alignment then counts correctly read marks as errors.

Definition. We take the reference lines of at least eight characters, longest first. For each line we look for the stretch of the output that matches it at under 40% character error rate, ignoring characters already claimed. We credit the marks that match and claim the stretch, so that no stretch credits two lines. The remaining lines are aligned globally, in reference order, against the output with all claimed characters removed. Without this second step, marks inside poorly read lines were lost, and our first version failed its check against the base model. Recall is the credited marks divided by the reference marks. Precision is the credited marks divided by all marks in the output, which penalizes surplus text. F1 is averaged over items by pooling counts.

Repeat-aware variant. In this variant, credit is computed after each loop run is reduced to one copy. A run is a unit of 4 to 400 characters that contains a letter and repeats exactly at least eight times, or six times for units of 50 characters or more. Precision still counts every mark in the full output, so the copies are penalized and cannot earn recall. We added the variant after a collaborator showed that the alignment matched 434 copies of one repeated unit to common syllables in the reference. When nothing loops, the two scores are identical.

Checks. Reading the same page twice in a row scores far below the better single read. The base model's recall stays within 0.3 points of the global alignment in every cell. Typhoon's full-page recall rises by 1.6 to 1.8 points when order is not charged, and charging order alone lowers its full-page mark F1 from 94.9 to 93.1 for the BQ prompt (95.3 to 93.7 for TC, both defined below). The base model is barely affected.

Data. We use the test split of ThaiOCRBench [1] on two tasks, full-page OCR and text recognition. A seeded calibration split, stratified by task and category, holds 69 full-page items and 109 text-recognition items. The rest of the benchmark is a locked split of 128 full-page and 224 text-recognition items. We did not open it until every rule in Sections IV and V was fixed, and we read it once, for the confirmation in Section VI. We report outputs and scores for each item and do not redistribute any benchmark image.

Models. We evaluate Typhoon OCR 1.5 (2B), its base model Qwen3-VL-2B, Pathumma-llm-vision 3.0, Qwen3-VL-4B, Qwen-SEA-LION-v4-4B-VL [22], DeepSeek-OCR-2 [23], and the 0.9B region recognizers PaddleOCR-VL-1.6 and wayu-paxa-ocr-zero (Wayu). We also apply the two region recognizers to full pages, which goes beyond their design. Sections IV and V report the first six models, and the last two enter in Section VI. DeepSeek-OCR-2 does not fit a T4 GPU in fp32, so all of its figures are fp16 and come from its own runner. Each checkpoint is pinned by revision, and we use the Apache-2.0-only revision of Typhoon. Revisions, seeds and run identifiers are recorded in the project repository.

Prompts and decoding. BQ is the benchmark's own question for each item, and TC is the prompt recommended on Typhoon's model card. Paddle and Wayu use their `OCR:` prompt, and Pathumma and DeepSeek-OCR-2 use the prompt on their model cards. Decoding is greedy with at most 3,072

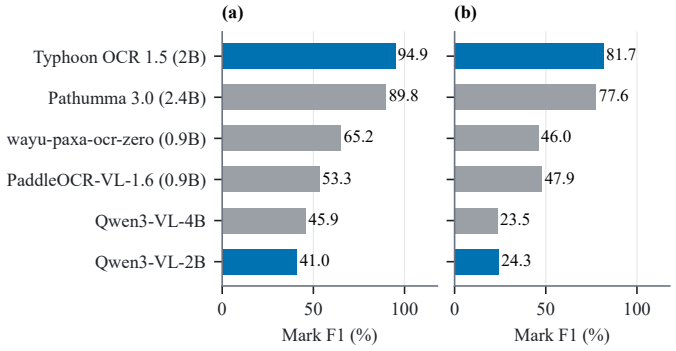


Fig. 1. Mark F1 on the calibration split, (a) full-page OCR and (b) text recognition. The prompt is BQ, except `OCR:` for the two 0.9B models and the card prompt for Pathumma.

new tokens, in fp16 on Kaggle T4 GPUs. Each result comes from one run. The greedy output and the output cut at a loop come from the same run, so the two are a paired comparison. Outputs are reduced to plain text before scoring. For Typhoon, BQ is the primary prompt and TC the secondary one.

Statistics. Intervals are 95% bootstrap intervals over items. Sections IV and V come from the calibration split. Their results are preliminary, and several decoding rules were written after that split had been seen, which we say where it applies. Section VI was registered before any locked output was scored, and uses 10,000 bootstrap resamples and one Holm correction.

IV. Where Marks Are Lost

Six models. On the calibration split, Typhoon scores a mark F1 of 94.9 on full pages and 81.7 on text recognition (Fig. 1 and Table I). Pathumma-llm-vision 3.0 [3], a second Thai fine-tune of the same base model, comes close. With its own card prompt it scores 89.8 and 77.6. Its difference from Typhoon is −5.1 points on full pages (interval −14.3 to +1.6) and −4.1 on text recognition (interval −15.1 to +8.0), so we cannot say that either model is better. Both are far above the base model, by 48.9 and 53.3 points. The other three checkpoints are well below Typhoon; Wayu, the best of them, is lower by 29.6 points on full pages (interval 19.2 to 39.9). Doubling the size of the base model changes its full-page F1 by only +5.0 (interval −4.0 to +14.1), and its tone error stays near 9.4 to 9.9%. Typhoon's median character error rate per page is 2.8%.

Two cautions apply to this comparison. The models are compared as released, with different numbers of visual tokens. And overlap with the benchmark cannot be excluded for the two best models: Pathumma's card reports ThaiOCRBench scores and does not name its 377K training samples, and Typhoon comes from the group that released ThaiOCRBench. We therefore call Typhoon the strongest model we measured, comparable to Pathumma, and nothing more.

On text recognition the two models fail differently. Pathumma finds more of the requested marks (recall higher by 14.3 points) but often transcribes the whole image instead of the requested span, which lowers its precision.

TABLE I

Mark F1, tone-mark error after a correctly read consonant, and share of outputs that reached the token limit (%). Pathumma uses its card prompt.

	Full-page			Text rec.	
Model	F1	tone err.	limit	F1	limit
Typhoon OCR 1.5	**94.9**	0.65	2.9	**81.7**	0.9
Pathumma 3.0	89.8	0.99	5.8	77.6	1.8
Wayu	65.2	5.27	26.1	46.0	20.2
PaddleOCR-VL-1.6	53.3	14.57	39.1	47.9	14.7
Qwen3-VL-4B	45.9	9.89	27.5	23.5	11.9
Qwen3-VL-2B	41.0	9.41	43.5	24.3	14.7

Loop copies earned recall. The repeat-aware score changes Typhoon by at most 0.2 points (F1 94.9 to 94.8, recall 96.1 in both). For the base model (BQ) it lowers full-page recall from 65.0 to 54.6, precision from 29.9 to 25.1 and F1 from 41.0 to 34.4. About a sixth of the marks that the order-free score credited to the base model came from loop copies. We read back every stripped run. All 126 had reached the token limit, the shortest contained 15 copies, and no ordinary text was removed.

Most lost marks are absent from the output. On full pages, Typhoon gets 6.2% (BQ) and 7.1% (TC) of all reference marks wrong. We assign each error automatically to one of four causes (Fig. 2). The line may be read in another order, meaning that some stretch of the output matches it at under 20% character error rate (for lines of at least eight characters). The whole line may be absent. A span may be missing inside a line that was kept. Or the mark may be misread. Absent lines account for 2.3% (BQ) and 2.9% (TC) of the marks, missing spans for 0.7% and 0.8%, and misreads for 1.5% and 2.4%. Reading order alone accounts for 1.6% and 0.9%, and this is not an error of reading. In the raw outputs, the absent lines are mostly text inside infographics and charts, inset advertisements, and titles or page furniture, where Typhoon writes an image placeholder. This agrees with the note on infographics by its authors [2]. The attribution is automatic, and we checked page images only for the six pages that hold most of the absent marks.

The same pattern holds for the other five models (Fig. 2). We apply the attribution to outputs cut at the first loop, because a looping output is many times longer than the reference and the alignment cannot be trusted on it. Whole lines absent plus missing spans cost 3.2% of the marks for Typhoon, 9.3% for Pathumma, and 29 to 44% for the other four models. A wrong mark after a correctly read consonant costs 0.4% for Typhoon, 0.8% for Pathumma, and 2.7 to 4.3% for the others. For every model, an absent line therefore costs between 7 and 12 times more marks than a misread mark. Absence from the output is what we measure. It is not the same as a failure to read, because a model may see a line and still leave it out. Telling the two apart would need, for example, a test in which the model reads the line alone as a crop, which we have not done on full pages. For the models that loop, a line that comes after a loop counts as absent, and we cannot separate lines lost to

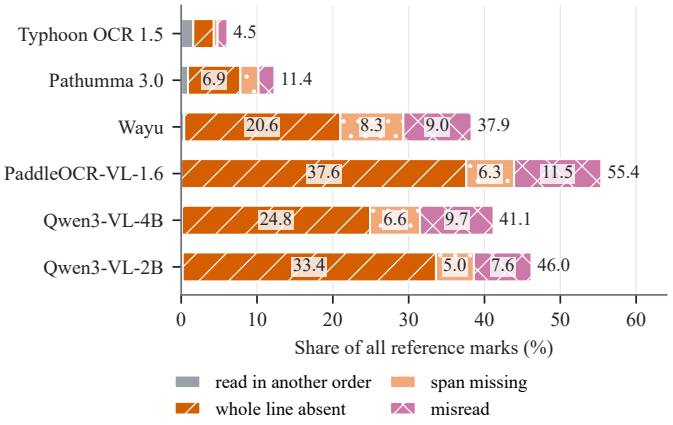


Fig. 2. Where full-page marks are lost, as a share of all reference marks, for outputs cut at the first loop. The bar totals exclude the grey segment, which is a difference in reading order and not an error.

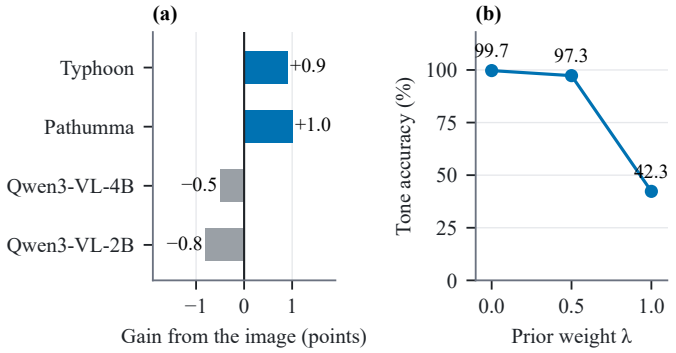


Fig. 3. (a) Gain in tone accuracy when the image is given, at the registered sites (calibration split). (b) Typhoon's tone accuracy when the no-image score is subtracted with weight λ.

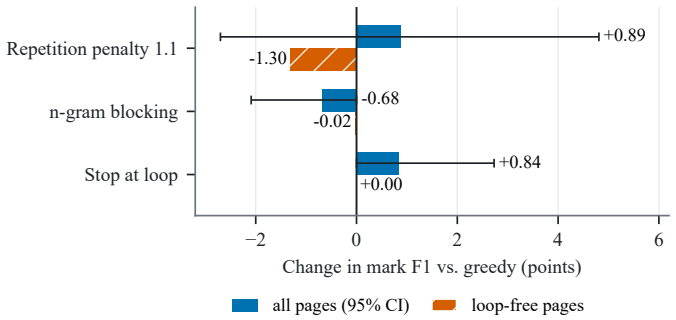


Fig. 4. Change in full-page mark F1 against greedy decoding (Typhoon, BQ). Blue bars cover all pages, with 95% intervals. Orange bars cover pages that do not loop.

the loop from lines the model would not have read anyway.

Given the right context, tone marks are nearly always right. In a forced-choice test, we give the model the correct text up to a mark and let it score the legal variants of that mark with and without the image (fp32, in-context tokenization). On full pages Typhoon chooses the correct tone mark 99.7% of the time with the image and 98.8% without it. For the base model the figures are 98.1% and 98.9%. The image therefore adds 0.9 points for Typhoon and removes 0.8 for the base model (Fig. 3a). This test gives the model correct context and is not free generation.

Two more models. Pathumma gives 99.7% with the image and 98.7% without it (gain +1.0), and 94.8% when the no-image score is subtracted at $\lambda = 0.5$. Qwen3-VL-4B knows the marks as well, at 98.7% with the image, and the image does not help it (99.1% without). The two differ when they write the page themselves. At sites where Pathumma read the base consonant correctly, 99.4% of its tone marks are right, against 99.8% for the test at the same sites. For Qwen3-VL-4B the figures are 91.3% and 99.1%: it misreads about one tone mark in eleven while generating, although the test shows that it knows the right one. The image adds a little for the two Thai fine-tunes and nothing for the general models, and we do not generalize beyond these four. For Qwen3-VL-4B, the finding that marks are mostly absent rather than misread holds in direction (on the calibration split, absent text costs 7.3 times more marks than these misreads) but not as a statement that marks are almost never misread.

Confidence. Typhoon's own token probabilities locate its mark errors. The lowest token probability inside a character cluster separates misread marks from correct ones with an AUROC of 0.918 (BQ) and 0.935 (TC) on the model's own greedy output.

V. Training-Free Decoding and Marks

Loops are rare but costly. Of 356 Typhoon outputs (both prompts, both tasks), 11 contain a loop. In each cell, two to five looping pages hold 68 to 76% of all surplus marks. Without them, full-page F1 for BQ would be 97.4 instead of 94.9. This figure is a ceiling obtained by removing those pages. It is not the result of any method.

Three ways to handle a loop. We compare three decoders for Typhoon with greedy decoding (Fig. 4, full pages, BQ).

- Repetition penalty 1.1, the setting used by Typhoon's own package, which applies the penalty of CTRL [18].
- n-gram blocking as in the DeepSeek-OCR code, which forbids repeating a 30-gram within 90 tokens.
- Stopping at a detected loop. When a unit repeats eight times, or six times for units of 50 characters or more, decoding stops at the first copy. Greedy decoding is prefix-deterministic, so we compute the cut offline from complete greedy outputs. We check a decode-time implementation against this cut in Section VI.

Repetition penalty. It changes nearly every output, and none of the loop-free full-page BQ outputs is identical to greedy. F1 on loop-free pages falls by 1.1 to 1.3 points. On text recognition (BQ) the tone error rises from 17.1% to 22.6%. Over all pages the effect cannot be told apart from zero (+0.89, interval −2.70 to +4.81). Thai text legitimately repeats syllables and tone marks, and a page-wide penalty may discourage them. We did not test this explanation.

n-gram blocking. It is safe where nothing loops: 92 to

100% of loop-free outputs are unchanged and the change in F1 is at most 0.05. It ends 6 of the 11 loops. In the others, the model escapes into near-repeats, for example a street name that drifts through three similar names. These near-repeats carry more marks, so surplus rises and full-page F1 ends 0.68 lower for BQ. We read a loop as an attractor of the decoder, so that banning an exact repeat moves the model to a neighboring one. This is an interpretation and has not been tested.

Stopping at a loop. It leaves every loop-free output unchanged, in every cell. Full-page mark F1 rises by 0.84 (BQ) and 0.67 (TC), and text-recognition F1 by 3.30 and 2.39. The stop also saves 11.2% of the generated tokens. Recall falls by at most 0.16 points, because stopping cannot recover text that the loop skipped. We wrote the rule after seeing the previous comparison. Each cell has only one to four looping outputs, so every interval includes zero.

The stop on other models. The other models loop far more often than Typhoon. On full pages, 18 to 30 of 69 outputs of the base model, Qwen3-VL-4B, Paddle and Wayu loop, against 2 for Typhoon and 4 for Pathumma. Applying the same offline stop to their greedy outputs raises mark F1 for every model on both tasks. No interval lies below zero, and seven of the twelve cells exclude zero. On full pages the gain is large where a model loops often, for example +24.4 for the base model and +10.8 for Wayu (interval 5.3 to 16.1), and small where it rarely loops: +0.8 for Typhoon and +3.0 for Pathumma. Section VI repeats the test on unseen items for all eight models.

The gain comes from precision and costs recall, up to 10.6 points on full pages. Part of that recall was credit earned by loop copies, and the rest is text that follows the loop and is cut. Across the twelve cells the stop changed 3 of 898 outputs that had ended on their own. Two of them are Paddle outputs that left a degenerate loop by themselves, and the cut dropped real page text after it. A repetition penalty of 1.05 alone, the value in Wayu's own example, does not help (+2.8, interval -3.6 to +9.6 on full pages, and -11.3, interval -22.5 to +4.5 on text recognition).

Stopping saves 34% of the generated tokens on Wayu's full pages. In this run, 72% of Wayu's full-page recall gap to Typhoon lies on the 18 pages where it loops, and no decoder we tested recovers it. We cannot say whether the loop is a cause or a symptom.

Subtracting the language prior removes correct marks. In the forced-choice test, subtracting the no-image score with weight λ lowers Typhoon's tone accuracy from 99.7% to 97.3% at $\lambda = 0.5$ and to 42.3% at $\lambda = 1$ (Fig. 3b). The base model falls to 84.7% at 0.5. In free generation, on a subset of 24 items, our M3ID-form contrast ended the loops of the base model (character error rate 1.12 to 0.21). It also raised the tone error on retained items from 3.7% to 7.6%, mostly by deleting tone marks (21 to 55 on 12 items). On Typhoon the change is small and concentrated in a few items. Both results point the same way. The prior for the tone mark is right, and removing it removes correct marks.

Speed. Speculative decoding with prompt lookup makes Typhoon 1.14 to 1.15 times faster (base model: 1.16 to 1.21 times) on pages that do not degenerate. The output is identical except at fp16 near-ties, and 88 to 93% of items are identical. The greedy reference reproduces our main outputs character for character on all 177 timed items. We have not measured the speed of the stop combined with speculative decoding, or the cost of checking for a loop at every token.

VI. Confirmation on the Locked Split

Everything above was developed on the calibration split. We then registered three tests, with their readings fixed in advance, and read the locked split once: 352 items (128 full-page, 224 text recognition) and eight models, namely the six above plus Qwen-SEA-LION-v4-4B-VL and DeepSeek-OCR-2. Prompts, greedy decoding and fp16 were as in Section III, and the cut at the first loop was applied offline to the greedy output of the same run. The tests were as follows.

- S1: for each model and task in which the model looped on at least 5% of calibration items, the stop raises mark F1. There are 13 such cells.
- S2: the stop changes at most 1% of the outputs that ended on their own, pooled over models and tasks.
- R1: for each model on full pages, the marks lost to absent lines and missing spans exceed three times those lost to misreading after a correctly read consonant, with the lower bound of the 95% interval above three.

One Holm correction covers all 22 tests (13 for S1, one for S2 and eight for R1). The intervals are page-bootstrap intervals with 10,000 resamples. The smallest one-sided p value this can give is 0.0001, so an adjusted value of 0.0021 is read as "at most 0.0021".

The stop. Mark F1 rises in 12 of the 13 eligible cells (Fig. 5 and Table II), by 4.8 to 35.6 points, with adjusted $p \leq 0.0036$. The thirteenth cell, Pathumma on full pages, does not pass. Its gain is +1.3 points (interval 0.0 to +3.5, adjusted $p = 0.13$). A post hoc rescoring with the two extraction defects described below removed gives +2.2 (interval 0.0 to +4.9, adjusted $p = 0.046$) for this cell, so the cell is sensitive to extraction; the registered result does not change. It was eligible on calibration by a small margin (4 of 69 pages looped) and looped on 4 of 128 locked pages (3.1%), the rate of the cells we did not test. Where a model rarely loops, the gain is small and not shown at this sample size. The criterion asked every eligible cell to pass, so S1 as registered is not met, and we name the cell. S2 passes: the stop changed 1 of 2,400 outputs that had ended on their own (0.04%, upper 95% bound 0.20%), a Typhoon output that left a real loop by itself.

Where marks are lost. R1 passes for all eight models (Table II). On full pages, absent text costs 4.7 times (PaddleOCR-VL, interval 3.2 to 6.7) to 15.8 times (Pathumma) more marks than a misread mark after a correctly read consonant. PaddleOCR-VL's lower bound is the closest to the threshold. This is a statement about these eight models on one benchmark. As before, absence from the output is what we measure, and it is not the same as a failure to read.

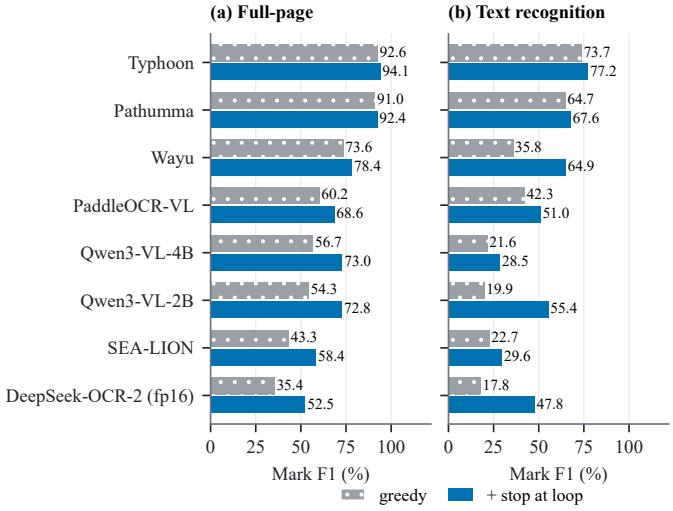


Fig. 5. Mark F1 on the locked split, with and without the stop at a detected loop, (a) full-page OCR and (b) text recognition. Prompts as registered. DeepSeek-OCR-2 is fp16 only.

TABLE II
LOCKED SPLIT. CHANGE IN MARK F1 WITH THE STOP (POINTS, 95% INTERVAL) AND R1, THE RATIO OF MARKS LOST TO ABSENT TEXT OVER MARKS LOST TO MISREADING (FULL PAGES). †NOT TESTED: THE MODEL LOOPED ON UNDER 5% OF CALIBRATION ITEMS. ‡REGISTERED TEST NOT PASSED.

Model	Full-page ΔF1	Text rec. ΔF1	R1
Typhoon OCR 1.5	+1.5 [−0.5, +4.1]†	+3.5 [0.0, +11.0]†	8.2 [5.5, 14.3]
Pathumma 3.0	+1.3 [0.0, +3.5]‡	+3.0 [0.0, +6.7]†	15.8 [7.5, 32.0]
Wayu	+4.8 [+1.8, +8.3]	+29.1 [+18.7, +38.0]	10.8 [7.1, 16.0]
PaddleOCR-VL-1.6	+8.4 [+4.5, +12.0]	+8.8 [+2.2, +14.6]	4.7 [3.2, 6.7]
Qwen3-VL-4B	+16.3 [+11.7, +20.3]	+7.0 [+4.2, +10.1]	6.1 [3.7, 9.2]
Qwen3-VL-2B	+18.4 [+14.3, +22.2]	+35.6 [+29.6, +40.9]	7.8 [5.0, 11.7]
SEA-LION v4 4B	+15.2 [+11.9, +18.4]	+6.9 [+3.8, +10.0]	7.8 [5.6, 10.4]
DeepSeek-OCR-2 (fp16)	+17.1 [+12.8, +21.5]	+30.0 [+21.8, +36.3]	7.8 [5.8, 10.5]

Ranking. Typhoon's mark F1 is higher than that of the six models other than Pathumma, on both tasks, with intervals above zero. Against Pathumma no difference is shown. With Pathumma's card prompt the difference is +1.5 points on full pages (interval −3.1 to +6.5) and +9.0 on text recognition (interval −3.6 to +21.3). With 128 and 224 items the half-widths are about 6 and 13 points, so this is not equivalence, and we do not call either model the best. Two extraction defects touch Pathumma's full-page outputs: an unclosed `<figure>` tag swallows the page text after it (nine outputs over the two prompts), and answers in Typhoon's JSON format keep their newlines as literal `\n` characters (30 outputs). Rescoring with both removed (post hoc) gives +2.4 (interval −2.6 to +7.8) with the card prompt on full pages, and the reading does not change. Typhoon's locked F1 (92.6 and 73.7) is below its calibration F1 (94.9 and 81.7). We tuned nothing on the locked items, so we expect such a drop on unseen pages, and we quote the locked figures as the headline.

Stopping live. The locked outputs were cut offline. To check the decode-time rule, we generated again on calibration items with the stop attached to generation, for all eight models: 185 items whose output the offline cut changes, and 72 sampled items that it leaves alone. The stopped output equals the offline cut character for character in 185 of 185 cases, and the rule did not fire on 72 of 72 outputs that the cut leaves alone. The looping outputs used 15.3% of the tokens that they used before. This check covers calibration items only, and we did not measure the time it saves.

VII. Discussion, Limitations and Conclusion

On the locked split of eight models, and on the calibration split of six, most of the marks lost are lost before the mark is reached: lines are absent from the output, spans are missing, or the model loops. Given the correct text, the tone mark is almost certain for the two Thai fine-tunes, and methods that remove the prior pay for it in tone marks. The cheap and safe way to deal with a loop is to stop it, and the gain is large where a model loops often. The picture is not the same for every model: Qwen3-VL-4B also misreads marks while it writes the page.

Scope. The study uses one benchmark and two tasks, and the models are open checkpoints studied as released, with different numbers of visual tokens. The two region recognizers are used on full pages beyond their design, and DeepSeek-OCR-2 ran in fp16 only, with its own runner. Typhoon comes from the group that released the benchmark, and Pathumma's training data are not named, so we cannot exclude overlap and do not rank the two. Few outputs loop for Typhoon and Pathumma, so most of their intervals include zero, and our claim concerns a mechanism rather than a rate.

Design. Sections IV and V come from the calibration split, and the stop rule, the repeat-aware score and the scoring were written after those data had been seen. Section VI is the confirmation. It was registered before any locked output was scored and read once, and its figures are the ones to quote. It supports the diagnosis and the stop for these eight models, and we claim nothing beyond them. S1 as registered was not met because of one cell. The stop was applied offline on the locked split, and the decode-time check covers calibration items only. On text recognition, several models have low precision because the items show a whole page while the question asks for part of it, which we did not test.

Interpretation. The attribution of errors is automatic. Absence from the output is not the same as a failure to read, and the attribution after a cut cannot tell a loop from a failure to read. Indirect evidence suggests that part of the absent text is text the model can read. In a separate pilot with one model on 69 questions about a boxed region, Typhoon found the boxed text in its answer for 7.2% of items when given the whole page (33.3% when the question did not state the coordinate system) and for 98.6% when given only the crop. That pilot asks about a region and not for a full-page transcription, covers one model, and does not say what share of our absent lines it explains. The stop sometimes cuts real text that follows a loop (3 of 898 outputs that had ended on their own on the calibration split, 1 of 2,400 on the locked split). The forced-choice test supplies correct context, so it shows what a model knows and not what it does when it generates. Our contrastive

variants are reimplementations tested on 24 items. Finally, we used a revision of Typhoon published under Apache-2.0 alone, while the vendor's current terms ask for written consent to competitive benchmarking.

Conclusion. A strong open Thai OCR model does not fail mainly at the tone mark. On a locked split, in all eight models we tested, the marks that are lost are lost mostly earlier, because the text is absent from the output: the model did not find or write it, or it looped. Given the right context, the two Thai fine-tunes choose the tone mark correctly about 99% of the time, although a general 4B model still misreads some marks as it writes. Stopping at a detected loop is a training-free step that raised mark F1 in 12 of the 13 registered cells and changed almost no output that had ended on its own. Removing the language prior is not a good way to recover marks. All results come from one benchmark.

Ethical and Licensing Notes

All models are open and used under their licenses (Apache-2.0 or MIT, with Typhoon at its Apache-2.0-only revision). ThaiOCRBench is released under CC-BY-SA-4.0. We cite its paper and revision, report outputs and scores, and do not redistribute its images.

References

[1] S. Nonesung, T. Jaknamon, S. Chaiophat, N. Nitarach, C. Wittayasakpan, W. Sirichotedumrong, A. Na-Thalang, and K. Pipatanakul, "ThaiOCRBench: A task-diverse benchmark for vision-language understanding in Thai," in *Proceedings of IJCNLP-AACL 2025 (Long Papers)*, 2025, pp. 1653–1675, arXiv:2511.04479. [Online]. Available: https://aclanthology.org/2025.ijcnlp-long.89

[2] S. Nonesung, N. Nitarach, T. Jaknamon, P. Taveekitworachai, and K. Pipatanakul, "Typhoon OCR: Open vision-language model for Thai document extraction," 2026. [Online]. Available: https://arxiv.org/abs/2601.14722

[3] NECTEC, "Pathumma-llm-vision-3.0.0-re," Hugging Face model card, 2026, revision 08bb11dcabf509ac3fa3ac0014ecd44eadb9729b. [Online]. Available: https://huggingface.co/nectec/Pathumma-llm-vision-3.0.0-re

[4] K. Pipatanakul, "How far can synthetic data take Thai OCR?" 2026. [Online]. Available: https://arxiv.org/abs/2609.03595

[5] S. Leng, H. Zhang, G. Chen, X. Li, S. Lu, C. Miao, and L. Bing, "Mitigating object hallucinations in large vision-language models through visual contrastive decoding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 13 872–13 882.

[6] A. Favero, L. Zancato, M. Trager, S. Choudhary, P. Perera, A. Achille, A. Swaminathan, and S. Soatto, "Multi-modal hallucination control by visual information grounding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, arXiv:2403.14003.

[7] S. Bai, Y. Cai, R. Chen *et al.*, "Qwen3-VL technical report," 2025. [Online]. Available: https://arxiv.org/abs/2511.21631

[8] C. Cui, T. Sun, S. Liang *et al.*, "PaddleOCR-VL: Boosting multilingual document parsing via a 0.9B ultra-compact vision-language model," 2025. [Online]. Available: https://arxiv.org/abs/2510.14528

[9] T. Do, D. P. Tran, A. Vo, and D. Kim, "Reference-based post-OCR processing with LLM for precise diacritic text in historical document recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 27, 2025, pp. 27 951–27 959.

[10] H. Elsafty, F. Aldabbas, and R. Sifa, "Arabic OCR in the age of multimodal models: A comprehensive comparative evaluation," in *2025 IEEE International Conference on Big Data*, 2025, pp. 5437–5442, lamarr Institute publication page.

[11] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi, "The curious case of neural text degeneration," in *International Conference on Learning Representations (ICLR)*, 2020, arXiv:1904.09751.

[12] L. Blecher, G. Cucurull, T. Scialom, and R. Stojnic, "Nougat: Neural optical understanding for academic documents," in *International Conference on Learning Representations (ICLR)*, 2024, arXiv:2308.13418.

[13] J. Poznanski, A. Rangapur, J. Borchardt, J. Dunkelberger, R. Huff, D. Lin, C. Wilhelm, K. Lo, and L. Soldaini, "olmOCR: Unlocking trillions of tokens in PDFs with vision language models," 2025. [Online]. Available: https://arxiv.org/abs/2502.18443

[14] PaddlePaddle, "PaddleX PaddleOCR-VL pipeline: `truncate_repetitive_content`," GitHub source code, 2025. [Online]. Available: https://github.com/PaddlePaddle/PaddleX/blob/ffb64904d23708863ff5b8da312a5cbd52a7f462/paddlex/inference/pipelines/paddleocr_vl/uilts.py

[15] DeepSeek-AI, "DeepSeek-OCR source code: `NoRepeatNGramLogitsProcessor`," GitHub, 2025. [Online]. Available: https://github.com/deepseek-ai/DeepSeek-OCR

[16] H. Wei, Y. Sun, and Y. Li, "DeepSeek-OCR: Contexts optical compression," 2025. [Online]. Available: https://arxiv.org/abs/2510.18234

[17] Z. Yang, J. Tang, Z. Li, P. Wang, J. Wan, H. Zhong, X. Liu, M. Yang, P. Wang, S. Bai, L. Jin, and J. Lin, "CC-OCR: A comprehensive and challenging OCR benchmark for evaluating large multimodal models in literacy," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 21 744–21 754.

[18] N. S. Keskar, B. McCann, L. R. Varshney, C. Xiong, and R. Socher, "CTRL: A conditional transformer language model for controllable generation," 2019.

[19] S. Liu, K. Zheng, and W. Chen, "Paying more attention to image: A training-free method for alleviating hallucination in LVLMs," in *European Conference on Computer Vision (ECCV)*, 2024.

[20] Y. Leviathan, M. Kalman, and Y. Matias, "Fast inference from transformers via speculative decoding," in *International Conference on Machine Learning (ICML)*, 2023, arXiv:2211.17192.

[21] A. Saxena, "Prompt lookup decoding," GitHub, 2023. [Online]. Available: https://github.com/apoorvumang/prompt-lookup-decoding

[22] AI Singapore, "Qwen-sea-lion-v4-4b-vl," Hugging Face model card, 2026, revision d0734aa1b8113fce90997a83c1e4afe2c37c6205. [Online]. Available: https://huggingface.co/aisingapore/Qwen-SEA-LION-v4-4B-VL

[23] H. Wei, Y. Sun, and Y. Li, "DeepSeek-OCR 2: Visual causal flow," 2026, model: deepseek-ai/DeepSeek-OCR-2, revision aaa02f3811945a91062062994c5c4a3f4c0af2b0. [Online]. Available: https://arxiv.org/abs/2601.20552